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import numpy as np

python_list = [1,2,3,4,5]
numpy_array = np.array(python_list)
print(numpy_array)
python_list2 = [[1,2,3],[4,5,6]]
numpy_array2 = np.array(python_list2)
print(numpy_array2)
print(numpy_array2.ndim)
arr = np.zeros((2,3))
print(arr)
arr = np.eye(3)
print(arr)
arr = np.arange(0,10,2)
print(arr)
arr = np.linspace(0,1,5)
print(arr)
arr = np.ones(4)
print(arr)

arr1 = np.array([1,2,3,4])
arr2 = np.array([6,7,8,9])
joined = np.concatenate([arr1,arr2])
print(joined)

a1 = np.array([1,2],[3,4])
a2 = np.array([5,6])
join = np.vstack((a1,a2))
print(join)

a = np.array([1],[2])
b = np.array([3],[4])
joining = np.hstack((a,b))
print(joining)

suruthi = np.array([1,2,3,4,5,6,7,8])
slipt_suruthi = np.array_split(suruthi,3)
print(slipt_suruthi)
print(slipt_suruthi[0])
print(slipt_suruthi[1])

a = np.array([[10,20],[30,40],[50,60],[70,80]])
print(np.split(a,2,axis=0))

arr = np.array([1,2,3,4,5,6,7,2])
search = np.where( arr == 2)
print(search)

arr = np.array([[1,2,3],
                [4,2,3]])
search = np.where( arr ==2)
print(search)

arr = np.array([1,4,5,25,68,9])
sort_array = np.sort(arr)
print(sort_array)

arr = np.array([10,20,30,40,50,60,70])
print(arr[0])
print(arr[-1])
print(arr[-3])
print(arr[1:4])
print(arr[2:])
print(arr[1:5:2])

arr1 = np.array([1,2,3,4,5])
for x in arr1:
    print(x)
arr2 = np.array([[1,2,3],[4,5,6]])
for x in arr2:
    print(x)
    for y in x:
        print(y)

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arr1 = np.array([10,20,30,40])
arr2 = arr1.copy()
arr2[0] = 100
print(arr1)
print(arr2)

rolls = np.random.randint(1,7,100)
print(rolls)

heights = np.random.normal(loc=170 , scale =10 , size =100)
print(heights)
count = np.sum(heights >= 160) & (heights <= 180)
print(count)

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```

[1 2 3 4 5]
[[1 2 3]
 [4 5 6]]
2
2
[[0. 0. 0.]
 [0. 0. 0.]
 [1. 0. 0.]
 [0. 1. 0.]
 [0. 0. 1.]]
[0 2 4 6 8]
[0. 0.25 0.5 0.75 1. ]
[1. 1. 1. 1.]
[1 2 3 4 6 7 8 9]
[[1 2]
 [3 4]
 [5 6]]
[[1 3]
 [2 4]]
[array([1, 2, 3]), array([4, 5, 6]), array([7, 8])]
[1 2 3]
[4 5 6]
[array([[10, 20],
        [30, 40]]), array([[50, 60],
        [70, 80]])]
(array([1, 7]),)
(array([0, 1]), array([1, 1]))
[ 1  4  5  9 25 68]
10
70
50
[20 30 40]
[30 40 50 60 70]
[20 40]
1
2
3
4
5
[1 2 3]
1
2
3
[4 5 6]
4
5
6
[10 20 30 40]
[100 20 30 40]
[2 1 6 1 3 4 1 5 2 6 6 2 2 3 6 3 3 6 1 4 4 5 5 5 5 5 4 1 4 5 5 6 6 1 5 6 3
 1 5 4 6 6 3 6 4 2 5 2 1 5 6 2 3 2 3 3 5 2 1 3 3 6 2 6 6 6 2 1 5 6 3 2 4 6
 6 6 2 4 3 5 3 3 2 5 2 4 5 4 1 2 4 4 5 6 3 6 1 5 6 3]
[166.10557942 159.04238628 173.40513791 164.42137995 177.1231679
 188.77618208 189.47327656 180.12382545 169.01633906 188.74589856
 164.24271245 187.92285001 180.9318616 171.66742783 182.49665661
 168.4088699 188.72925404 173.37141932 164.33427011 168.0988449
 177.62279565 178.8951535 170.1812864 189.85917157 164.46445177
 184.11708599 167.46880437 166.3896464 164.67634335 164.59449462
 162.99904295 169.23853461 176.80954464 173.15037346 178.47371833]

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